

Research Paper Replication Summary (Phase 1)

Course: DA351 — Explainable AI (XAI) | **Project:** Phase 1 Deliverable: Research Paper Replication (20 Marks)

Paper: Do Models Explain Themselves? Counterfactual Simulatability of Natural Language Explanations (Chen et al., EMNLP 2023)

Evaluated Dataset: StrategyQA Test Benchmark (30 questions × 10 counterfactuals × 4 setups = 1,200 evaluations)

1. Objective and Problem Statement

Large Language Models frequently produce fluent explanations that sound persuasive to humans but fail to reflect how the model actually makes decisions. Evaluating explanations based solely on human plausibility is problematic because models can hallucinate convincing rationales for wrong answers. Chen et al. (EMNLP 2023) introduced **Counterfactual Simulatability** to address this issue: an explanation is considered faithful only if it enables an observer to accurately predict how the model will answer new, related counterfactual questions. In Phase 1, our goal was to replicate this evaluation framework and reproduce its core empirical benchmarks on the StrategyQA dataset.

2. Methodological Architecture

The replication follows a four-step evaluation pipeline:

- **Explanation Generation:** The evaluated model answers a multi-hop question under two distinct paradigms: Chain-of-Thought (step-by-step reasoning before the answer) and Post-Hoc (justification generated after the answer).
- **Counterfactual Generation (SimQG):** A generator model takes the original question, answer, and explanation, synthesizing 10 follow-up yes/no questions whose answers can be inferred directly from the explanation.
- **Simulation (SimQA):** A simulation agent reads the original explanation and new counterfactual questions. If the explanation contains enough information, it predicts the model's answer; if not, it outputs "cannot guess" ($\perp$), filtering out invalid questions.
- **Scoring:** We compute Simulation Precision (fraction of correct predictions on answerable counterfactuals) and Simulation Generality (diversity among generated questions using BLEU, Sentence-BERT Cosine distance, and Jaccard token overlap).

3. Implementation and Engineering Highlights

- **Evaluated Models:** We used open-weight models via Groq Cloud: [openai/gpt-oss-20b](#) (representing small base models, proxy for GPT-3.5) and [openai/gpt-oss-120b](#) (representing large flagship models, proxy for GPT-4 and the simulator).

- **Caching Layer:** Built an SQLite database cache (`groq_cache.sqlite`) that stores past responses by prompt hash, ensuring zero duplicate API calls and deterministic repeatability.
- **Batching Optimization:** Grouped all 10 counterfactual questions into a single structured query, reducing API round-trips by over 80% and allowing all 1,200 evaluations across 4 conditions to complete in 40 minutes.

4. Key Replicated Results

- **Scale Improves Simulatability (Table 4):** On Chain-of-Thought reasoning, precision increased from 74.94% on 20B to 85.78% on 120B (+10.84% gain). This reproduces the paper's core finding that larger models generate explanations with stronger logical entailment.
- **Benchmark Accuracy:** Averaged across all 4 conditions, our replicated simulation precision reached 79.57%, matching the published paper average of 79.80% within 0.23%.
- **High Counterfactual Diversity (Table 5):** Generated questions achieved high diversity across BLEU (87.25%), Cosine (33.38%), and Jaccard (81.93%), confirming that counterfactuals test diverse reasoning angles rather than trivial rephrasings.
- **Explanation Faithfulness vs. Task Accuracy (Table 6):** High base task accuracy did not directly determine simulation precision, validating that simulatability evaluates explanation transparency rather than simple memorization.